

EXPLORATORY DATA ANALYSIS OF GOOGLE CLOUD RESOURCE CONSUMPTION AND COST EFFICIENCY

Resource Utilization, Infrastructure
Efficiency & Cloud Cost Patterns



Google Cloud

TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

PRESENTED BY

Tejas Jadhav

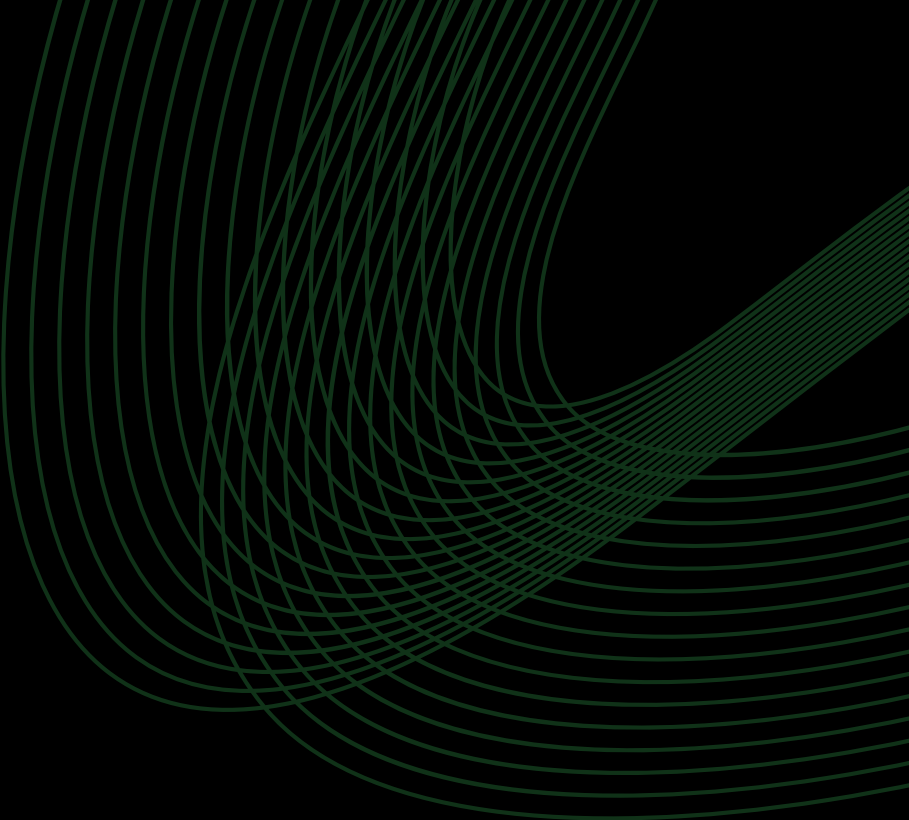


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PROBLEM STATEMENT & OBJECTIVE

Problem Statement:

Managing Google Cloud Platform resources across multiple services, regions, and workloads is complex, with diverse usage patterns and cost drivers. This makes it challenging to identify the primary factors influencing cloud expenditure, resource efficiency, and opportunities for targeted cost optimization.

Objective:

To analyze the Google Cloud Platform usage and cost dataset to uncover patterns, insights, and trends that can help businesses make data-driven decisions around resource efficiency, cost optimization, and cloud expenditure management.



EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01

Data Collection

- Gathered the Google Cloud Platform Dataset from Kaggle

02

Data Understanding & Anomaly Detection

- Looked at data distributions
- Checked for missing values, outliers and unusual patterns

03

Data Cleaning & Treatment

- Fixed incorrect values
- Standardized formats for consistency

04

Exploratory Analysis

- Univariate Analysis: e.g Top Services, Region/Zone, Usage Start Date etc
- Bivariate: CPU Utilization vs Total Cost, Average Total Cost by Service Name
- Multivariate Analysis: Total Cost vs Service Name vs Region

05

Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

KEY QUESTIONS EXPLORED

- How do resource type, configuration, and instance size influence overall GCP spend and cost efficiency?
- Which high-cost Resource IDs disproportionately contribute to cloud expenditure, and where can targeted optimization yield the largest savings?
- How do CPU, memory, and network utilization patterns correlate with actual spend, and which workloads are over-provisioned?
- To what extent do regional deployment choices impact cost, and are there opportunities to optimize for latency, compliance, and spend?
- How do network traffic patterns (inbound/outbound) affect total cost, and what strategies could reduce egress and routing charges?
- Which services (e.g., Cloud Build, Cloud Storage, Dataproc) are driving the fastest cost growth, and where is spend declining due to optimization?
- How does usage volume versus pricing tiers influence cost volatility, and where can standardization or rightsizing improve predictability?
- What role does workload-specific instance selection (compute vs memory optimized) play in maximizing infrastructure efficiency?
- Which optimization strategies (rightsizing, autoscaling, traffic routing, or regional adjustments) are likely to deliver the highest ROI for cost reduction?



DATA OVERVIEW

The dataset provides insights into Google Cloud Platform resource usage and costs, covering service-level consumption, instance configurations, regional deployment, network traffic, and workload characteristics. It also captures utilization metrics, cost trends, and performance indicators that are relevant for understanding efficiency, optimization opportunities, and drivers of cloud expenditure.

Data Source: Kaggle

Dataset Size

1,000

Records

15

Features

Purchase Diversity

12

Regions

25

Services

3

Usage Units



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Resource ID	Categorical	Unique identifier for the GCP resource being used
Service Name	Categorical	Name of the Google Cloud service associated with the usage
Usage Quantity	Categorical	Amount of the resource consumed during the billing period (e.g., VM hours, GB of storage)
Usage Unit	Categorical	Device type used during the session (e.g., desktop, mobile, tablet)
Region/Zone	Categorical	Geographic region or zone where the resource is located
CPU Utilization (%)	Numerical (Continuous)	Average CPU usage percentage for the resource over the billing period
Memory Utilization (%)	Numerical (Continuous)	Average memory usage percentage for the resource over the billing period
Network Inbound Data (Bytes)	Numerical (Discrete)	Amount of incoming network traffic in bytes
Network Outbound Data (Bytes)	Numerical (Continuous)	Amount of outgoing network traffic in bytes
Usage Start Date	Date / Time	Start date of the usage period
Usage End Date	Date / Time	End date of the usage period
Cost per Quantity (\$)	Numerical (Continuous)	Price of a single unit of the resource
Unrounded Cost (\$)	Numerical (Continuous)	Raw cost calculated before rounding
Rounded Cost (\$)	Numerical (Discrete)	Cost rounded to the nearest cent
Total Cost (INR)	Numerical (Discrete)	Total cost of the resource usage converted to Indian Rupees

DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

DATA ANOMALIES

- Usage Start Date and Usage End Date were stored as str and should be converted to datetime for proper time-based analysis.



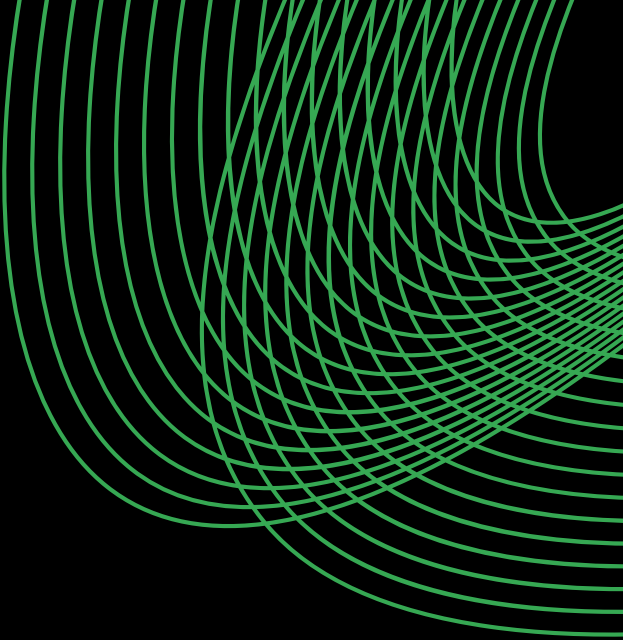
DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

DATA CLEANING SUMMARY

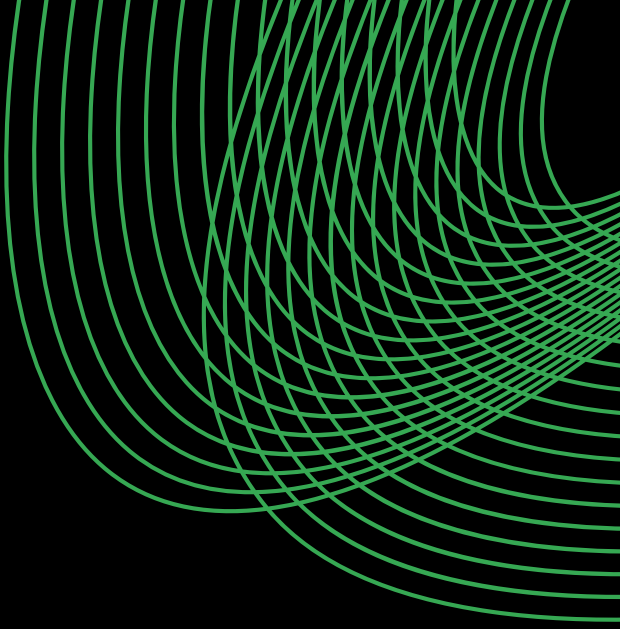
- Categorical fields such as Resource ID, Service Name, Usage Unit, and Region/Zone were converted to categorical data types.
- Usage Start Date and Usage End Date were standardized to datetime format to support efficient and accurate time-based analysis.
- To make cost analysis globally comparable, Total Cost was converted from Indian Rupees (INR) to US Dollars (USD) using the applicable exchange rate, facilitating consistent reporting and cross-regional insights.





INSIGHTS





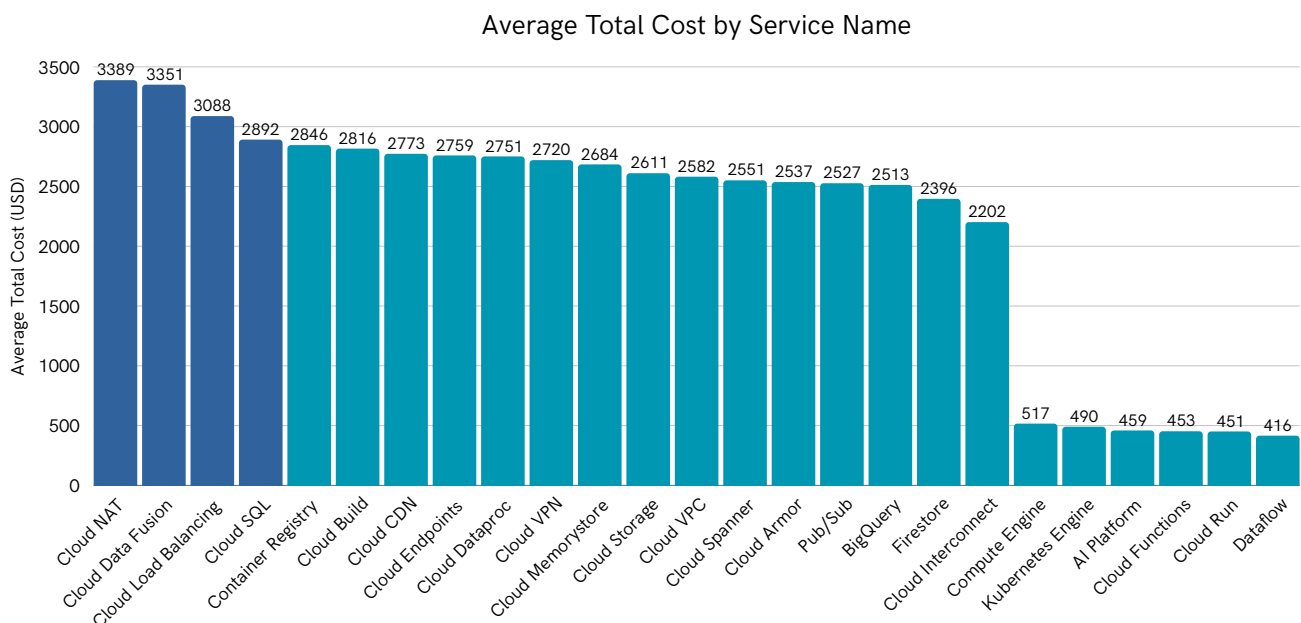
Cloud Spend Overview



Network-heavy services dominate cloud spend, outweighing compute and serverless workloads

44.47%

User Traffic is from the United States



Key observations

- Cloud NAT and Cloud Load Balancing are the top cost drivers, significantly exceeding all other services
- Core networking and data movement services (NAT, Load Balancing, CDN) collectively account for a much larger share of spend than compute-adjacent services like Kubernetes Engine or Cloud Functions.

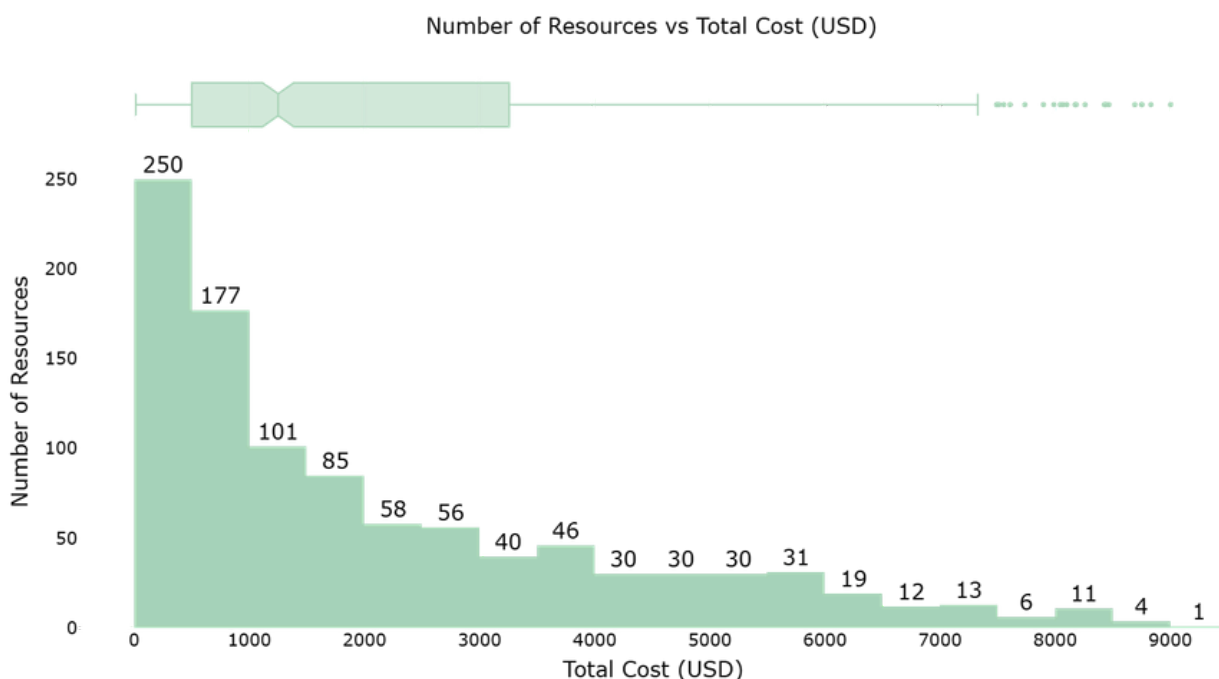
Business Insights

- Cloud spend is being driven more by traffic routing and data transfer than by raw compute usage, indicating potential inefficiencies in network architecture.
- Cost optimization efforts should prioritize network egress, routing design, and traffic patterns within Google Cloud Platform, as marginal improvements here can unlock outsized savings compared to tuning smaller compute workloads.

Cloud spend is highly right-skewed, with a small number of resources driving extreme costs

58.20%

of User Sessions come from Desktops



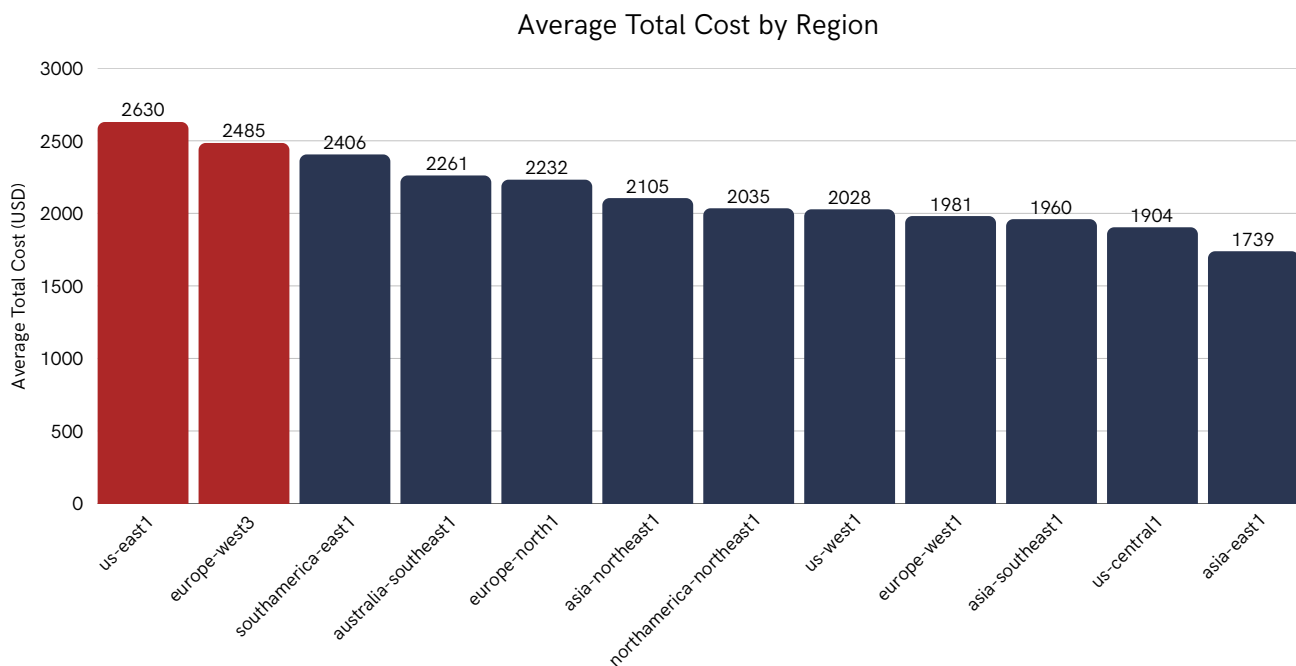
Key observations

- Most resources incur relatively low monthly costs, clustering at the lower end of the spend distribution.
- A long right tail shows a limited number of resources with very high costs, extending far beyond the typical spend range.

Business Insights

- Overall cloud expenditure is disproportionately influenced by a small subset of high-cost resources, making them the primary leverage points for savings.
- Targeted optimization, rightsizing, or architectural review of these high-spend resources on Google Cloud Platform is likely to deliver far greater cost reductions than broad, across-the-board tuning.

Cloud resources are broadly distributed across regions, with mild concentration in Asia and Europe



Key observations

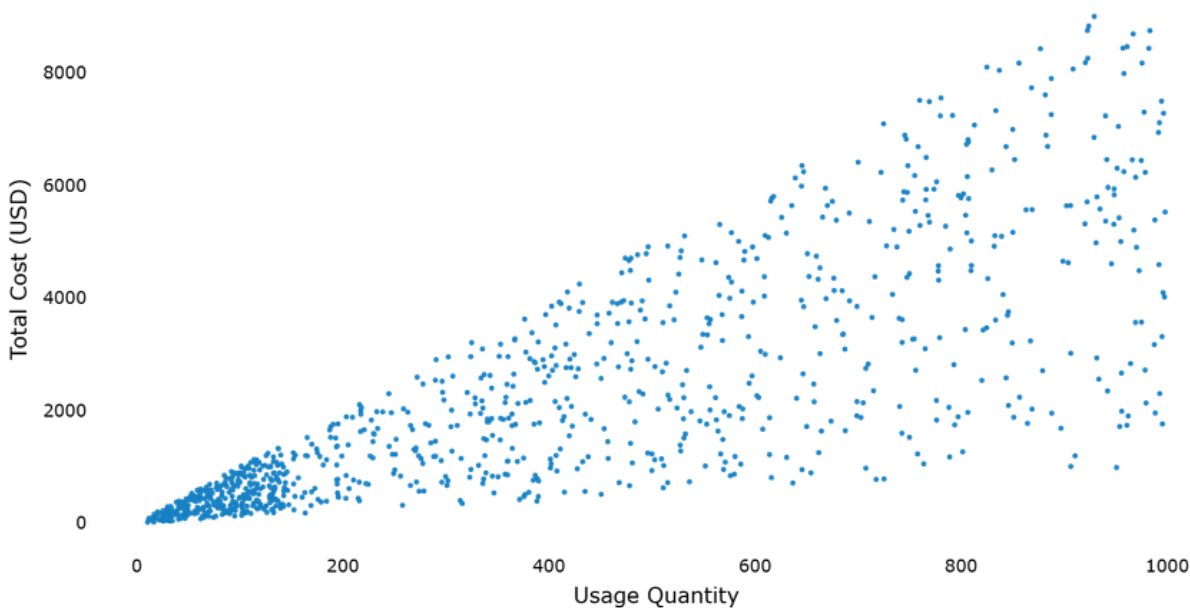
- asia-east1 and europe-west1 host the highest number of resources, but no single region overwhelmingly dominates deployment.
- Resource counts across regions remain relatively balanced, with only a gradual drop-off toward lower-ranked zones.

Business Insights

- The regional footprint suggests a geographically diversified infrastructure, likely supporting latency, availability, or regulatory needs.
- Since deployment is not heavily concentrated in one region, cost optimization should focus on service-level and usage efficiency rather than large-scale regional consolidation within Google Cloud Platform.

Cloud costs scale with usage volume, but pricing inefficiencies widen at higher consumption levels

Higher Consumption Directly Drives Cost Escalation



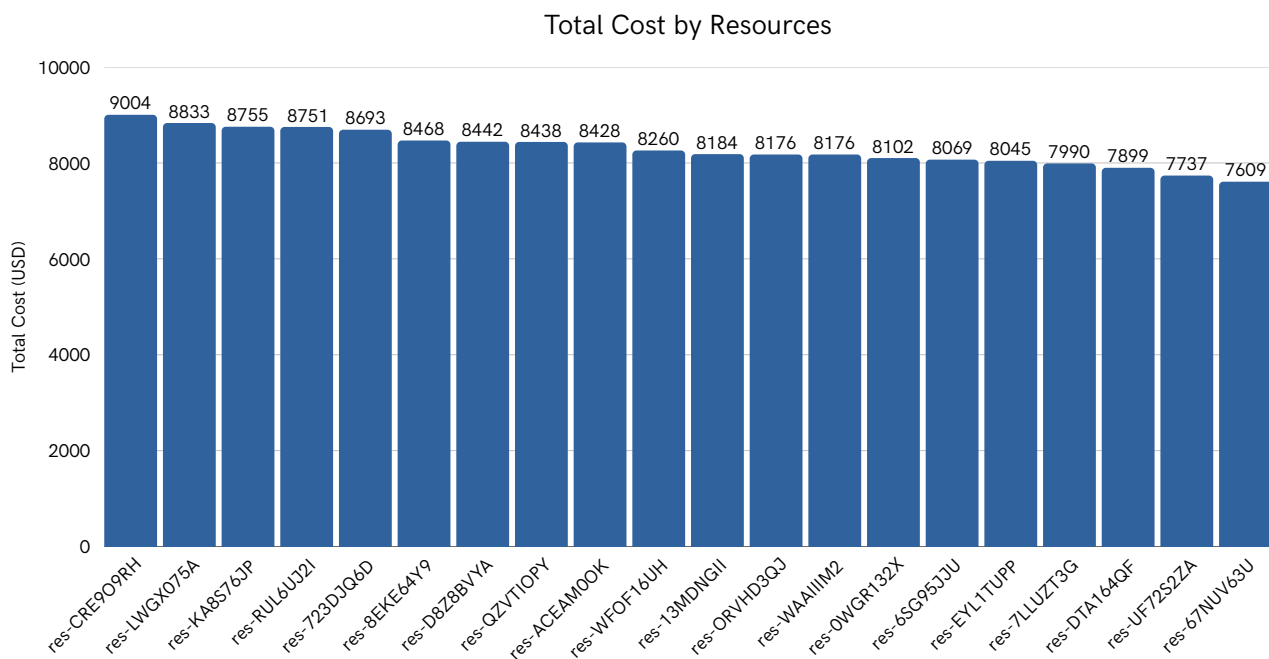
Key observations

- Total cost increases with usage quantity, showing a strong positive relationship between consumption and spend.
- Cost dispersion widens significantly at higher usage levels, with similar usage quantities resulting in very different total costs.

Business Insights

- Higher usage does not translate into linear cost growth, indicating pricing tier effects, service mix differences, or inefficient resource configurations.
- Standardizing high-usage resources and reviewing unit pricing models on Google Cloud Platform can reduce cost volatility and prevent excessive spend at scale.

A small set of individual resources accounts for a disproportionately high share of cloud spend

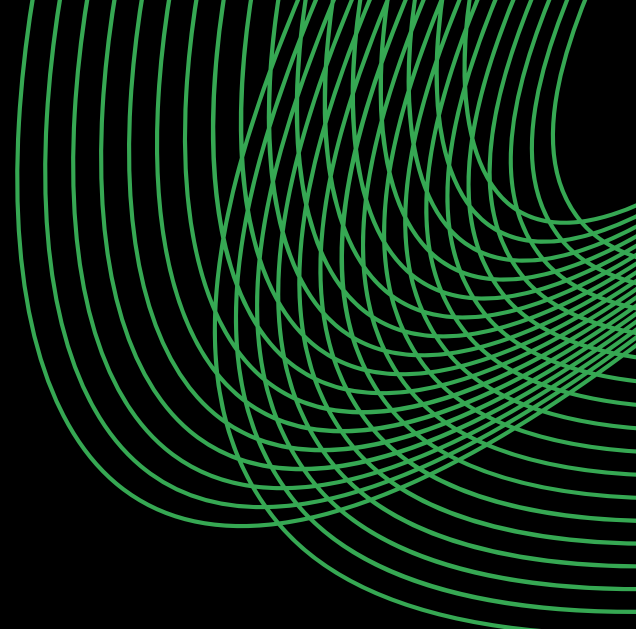


Key observations

- The top 10-15 resources each incur exceptionally high total costs, far exceeding the typical resource-level spend.
- Cost levels among these top resources are tightly clustered, indicating multiple consistently expensive resources rather than a single outlier.

Business Insights

- Cloud expenditure is highly concentrated at the resource ID level, making these assets the most impactful targets for optimization.
- Conducting detailed evaluations of these high-cost resources, including their usage levels, instance sizing, and service configurations, can reveal meaningful cost optimization opportunities within Google Cloud Platform while maintaining the stability of the overall infrastructure.



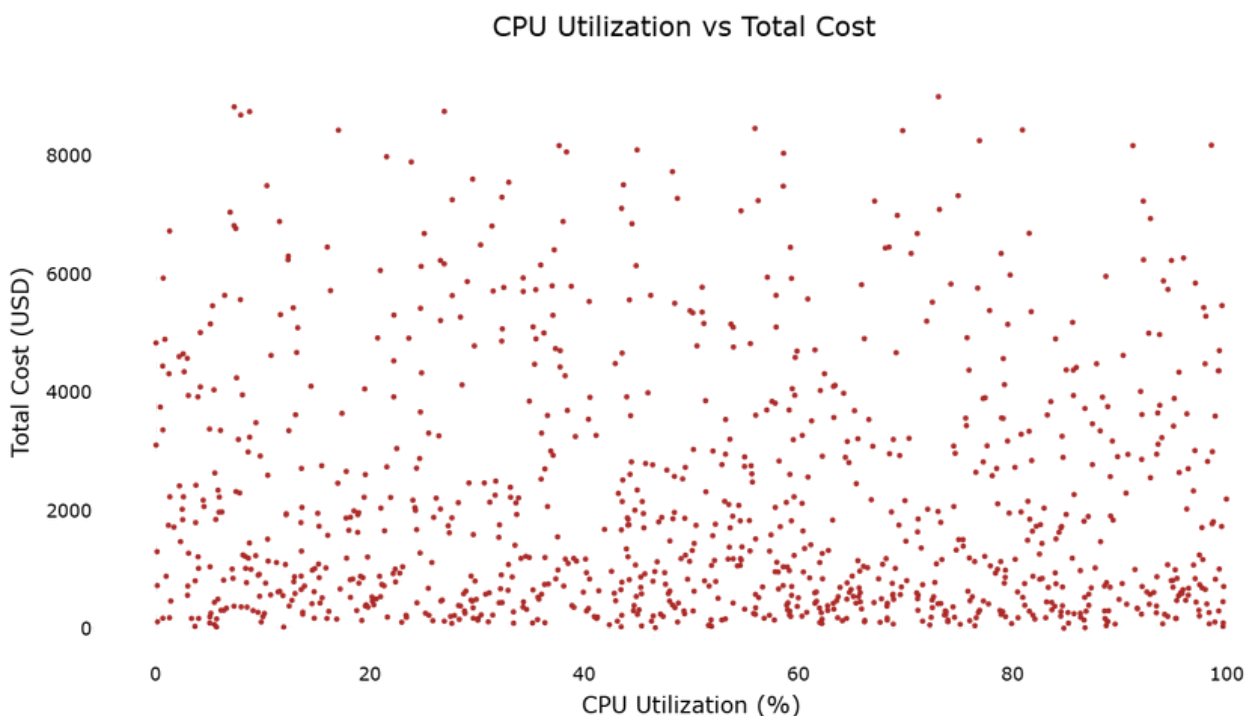
Product Performance & Preferences



CPU Utilization Has Weak Correlation with Total Cost, Indicating Potential Over-Provisioning

38.85%

Of the Interactions is driven by Apparel



Key observations

- Total cost is widely distributed across all CPU utilization levels (0–100%), with no clear upward trend as utilization increases.
- Several high-cost resources exist even at low CPU utilization levels, indicating that high spending is not necessarily driven by high compute usage.

Business Insights

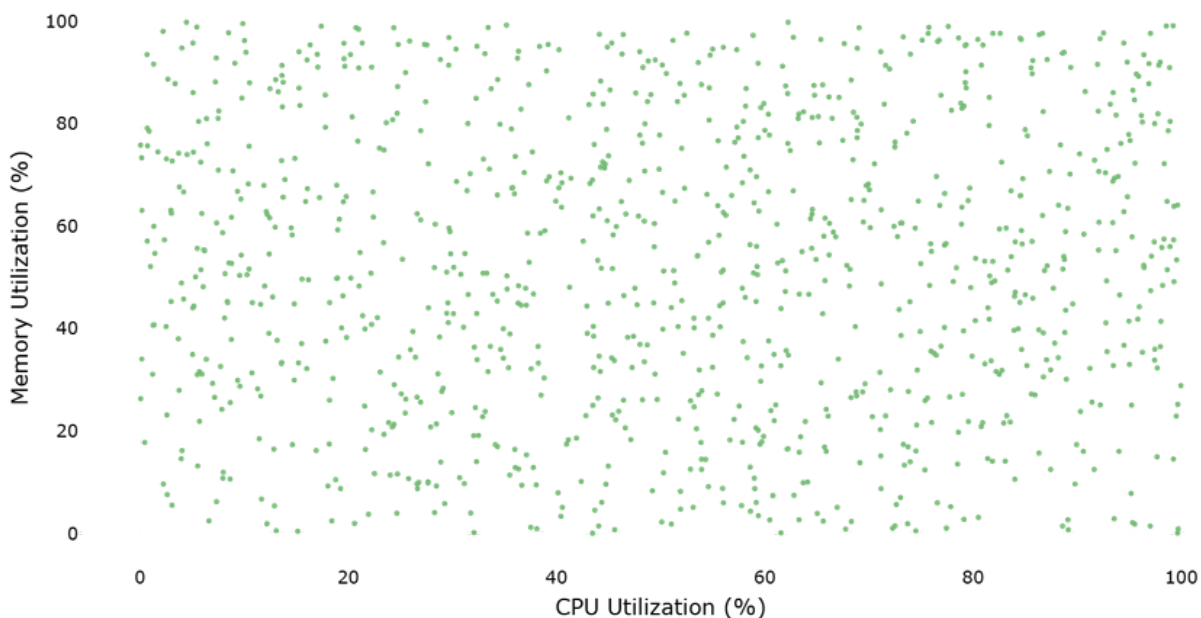
- Costs are likely driven more by resource allocation (instance size/type) rather than actual CPU consumption, suggesting over-provisioned infrastructure.
- Implement rightsizing strategies and enable autoscaling to align compute capacity with actual workload demand, reducing unnecessary cloud spend.

CPU and Memory Utilization Show Low Alignment, Indicating Imbalanced Workloads

76%

Of Users search for Google related products

CPU Utilization vs Memory Utilization



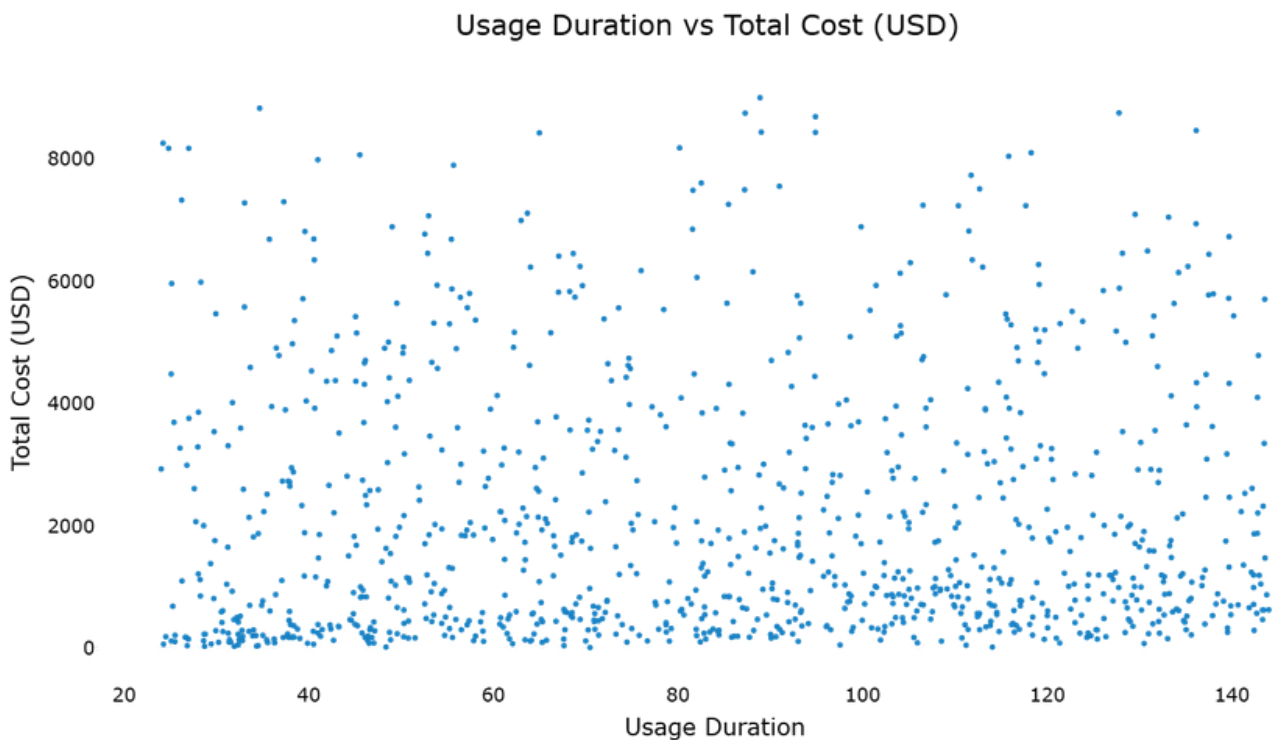
Key observations

- Memory utilization is widely spread across all CPU utilization levels, showing no strong linear relationship between CPU and memory usage.
- High CPU utilization does not consistently correspond with high memory utilization, suggesting uneven resource consumption patterns.

Business Insights

- Workloads may not be optimally matched to instance types, leading to inefficient resource allocation (e.g., memory-heavy instances with low memory usage).
- Adopting workload-specific instance selection (compute-optimized vs memory-optimized) can improve infrastructure efficiency and reduce unnecessary provisioning costs.

Extended Usage Duration Does Not Proportionally Increase Total Cloud Cost



Key observations

- Total cost remains widely distributed across all usage durations, with no clear upward trend as duration increases.
- Several short-duration resources incur high costs, while many long-running resources maintain relatively low total costs.

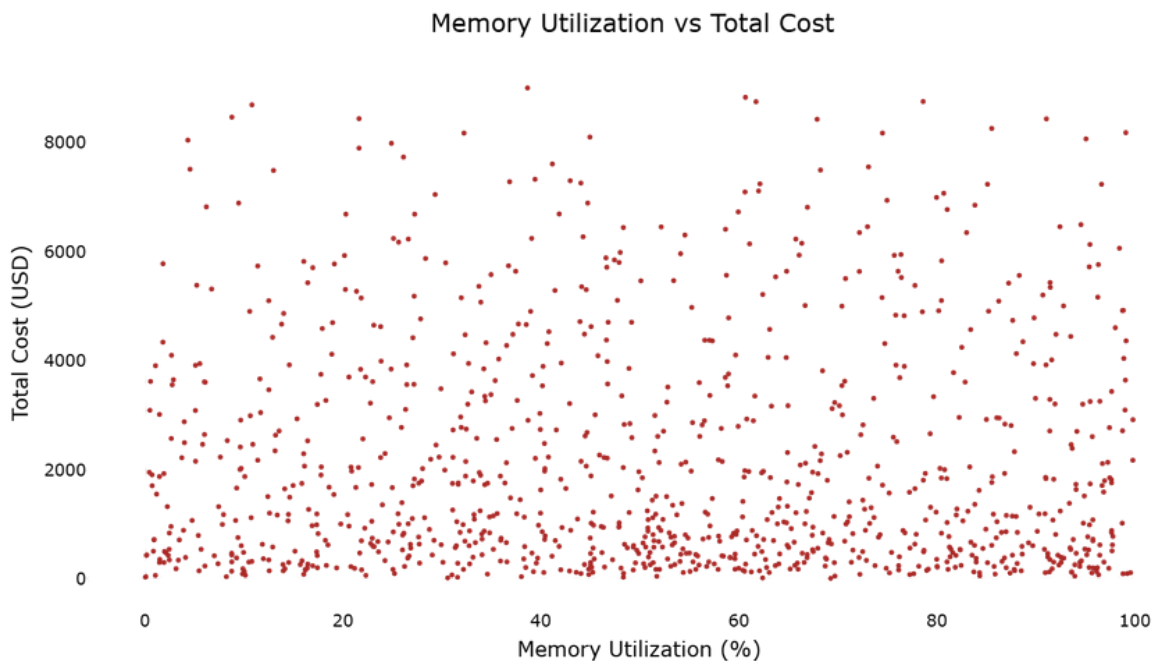
Business Insights

- Cost is driven more by resource type and configuration than by runtime length alone, indicating that high-spec resources significantly impact spend even for shorter periods.
- Focus cost optimization efforts on high-cost resource configurations (instance size, storage class, premium tiers) rather than only targeting long-running workloads.

No Clear Correlation Between Memory Utilization and Total Cost on GCP Resources

\$22

Typical Price (Median)



Key observations

- Total costs are widely spread across all levels of memory utilization, indicating no consistent pattern that higher memory usage leads to higher cost.
- Many resources incur low to moderate costs even when memory utilization is at extreme low or high percentages, showing cost drivers may be independent of memory load.

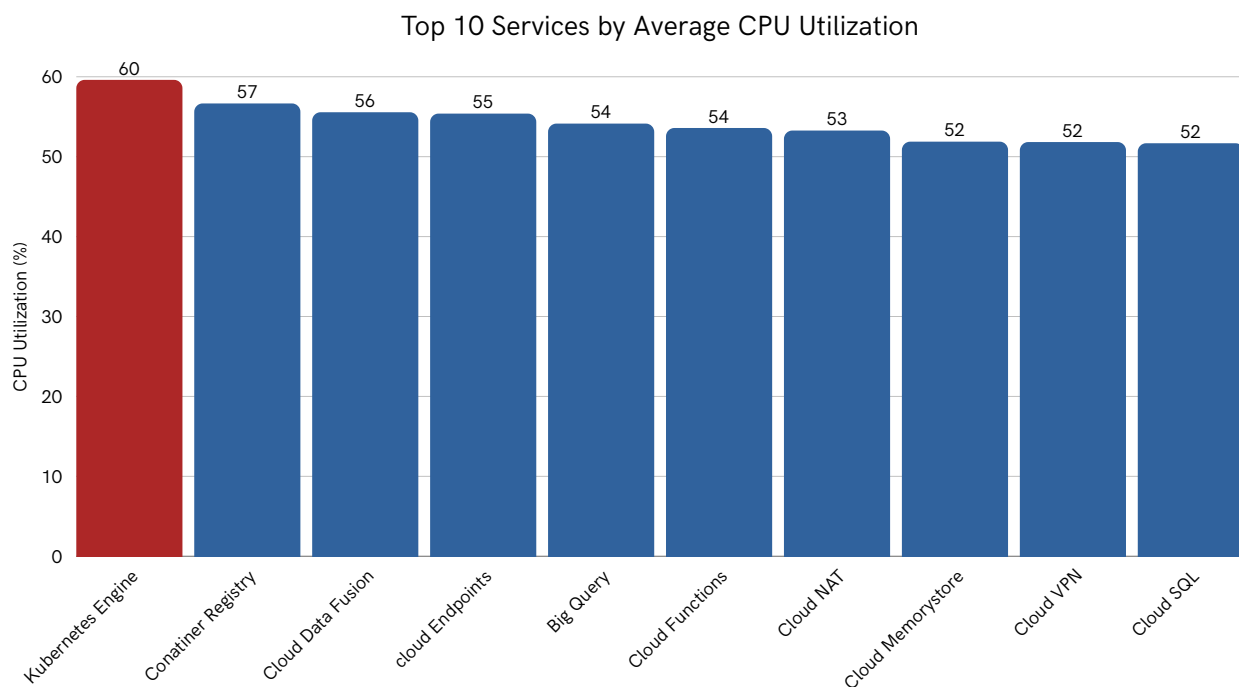
Business Insights

- Memory utilization alone is not a reliable predictor of cost; cost optimization efforts should focus on other factors like resource type, region, or usage quantity.
- Resources with very low memory utilization but significant cost may represent opportunities to downsize or reconfigure to reduce spend without impacting performance.

Top GCP Services Exhibit Moderate to High Average CPU Utilization

\$22

Typical Price (Median)

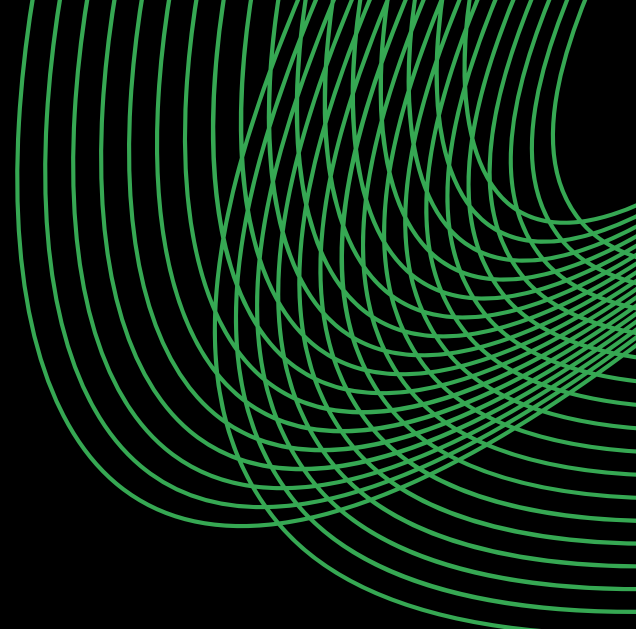


Key observations

- Kubernetes Engine leads CPU utilization with an average of approximately 60%, indicating heavy compute workload on containerized infrastructure.
- All top 10 services show CPU utilization above 50%, reflecting consistent resource consumption across diverse GCP services such as Container Registry, BigQuery, and Cloud Functions.

Business Insights

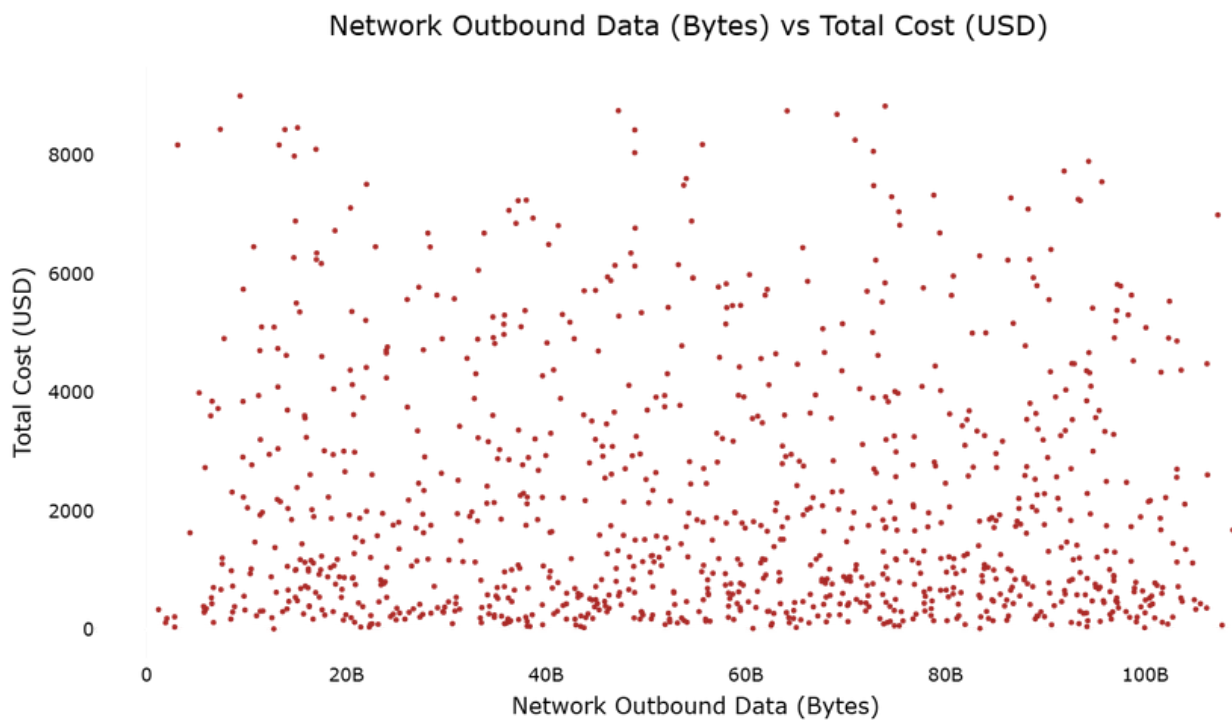
- High CPU utilization in Kubernetes Engine suggests optimizing container workloads and autoscaling policies could improve performance and cost efficiency.
- Consistently moderate CPU usage across multiple services indicates potential for resource tuning or rightsizing to avoid overprovisioning and reduce unnecessary costs.



Network & Hidden Cost Drivers



No Strong Correlation Between Network Outbound Data and Total Cost on GCP Resources



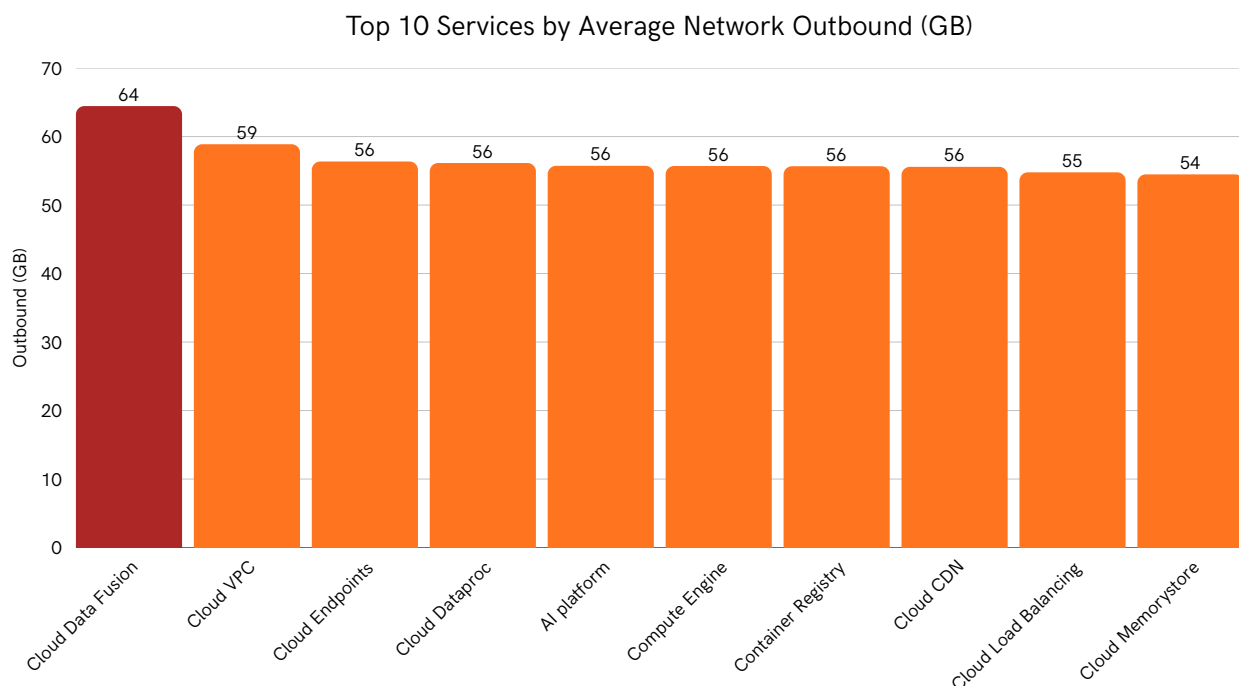
Key observations

- Total cost values are spread widely across all levels of outbound network data, indicating high data transfer does not necessarily lead to higher costs.
- Several resources with low outbound data usage still incur significant costs, suggesting that factors other than network traffic heavily influence overall spend.

Business Insights

- Network outbound data alone is not a major cost driver, so optimization efforts should focus on other cost components like compute or storage resources.
- Resources with low data transfer but high costs could be reviewed for inefficiencies, underutilization, or potential rightsizing to improve cost efficiency.

High Network Egress Concentration in Cloud Data Fusion and Endpoint Services



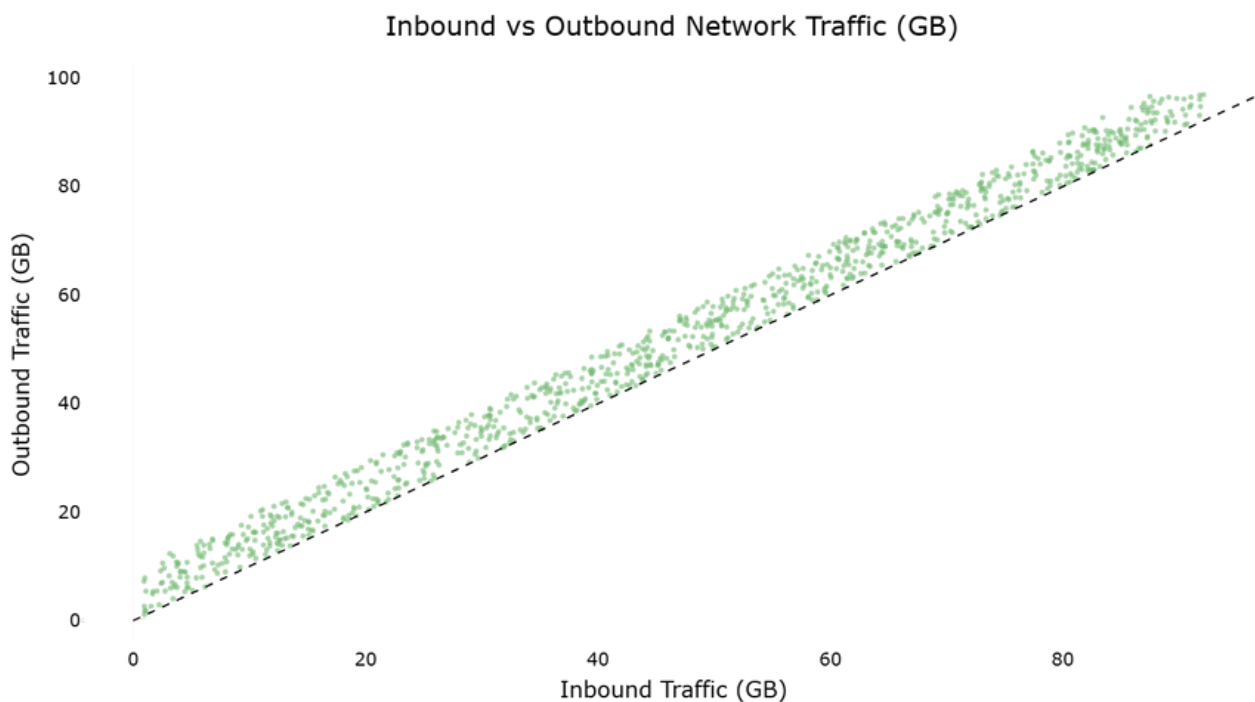
Key observations

- Cloud Data Fusion records the highest average outbound network usage (~64 GB), significantly exceeding other services clustered between ~46–58 GB.
- Data-intensive services like Cloud Endpoints, AI Platform, and Container Registry also show consistently high outbound traffic, indicating heavy external data movement across core processing services.

Business Insights

- High outbound traffic from Cloud Data Fusion suggests potential cost exposure to network egress charges; optimizing data transfer architecture or implementing regional data localization could reduce bandwidth costs.
- Services with sustained high outbound usage should be evaluated for caching, compression, or VPC peering strategies to minimize external traffic and improve cost efficiency.

Strong Positive Correlation Between Network Inbound and Outbound Traffic

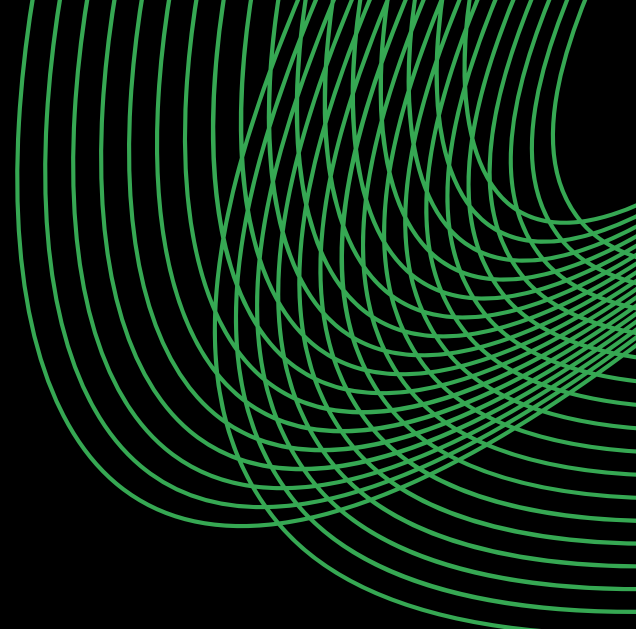


Key observations

- There is a near-linear positive relationship between inbound and outbound traffic, indicating that as incoming data increases, outgoing data scales proportionally.
- The majority of data points closely follow the trend line, suggesting consistent bidirectional traffic behavior across services without extreme outliers.

Business Insights

- The proportional scaling of inbound and outbound traffic suggests predictable network cost growth; forecasting models can reliably estimate future egress costs based on inbound trends.
- Since outbound traffic often incurs higher charges, optimizing data processing to reduce unnecessary external responses (e.g., filtering or aggregation before transmission) could significantly lower network-related expenses.

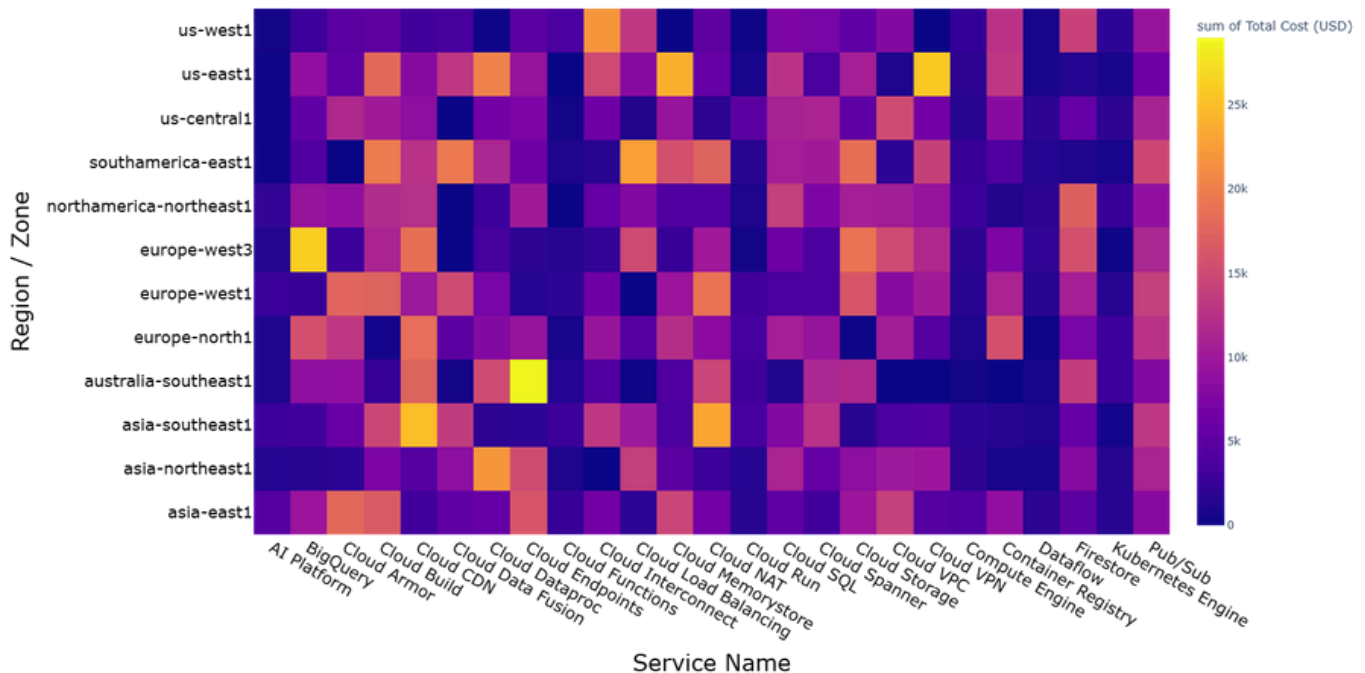


Service & Regional Strategy



A Small Set of Services and Regions Contribute Disproportionately to Total Cloud Costs

Spending Concentrates Around Specific Services in Key Regions



Key observations

- Regions such as us-east1, europe-west3, and asia-south1 show noticeably higher total costs across multiple services, indicating geographic concentration of cloud spending.
- Services like Cloud Data Fusion, Cloud CDN, Cloud SQL, and AI Platform display recurring high-cost intensity across several regions, suggesting they are primary cost drivers.

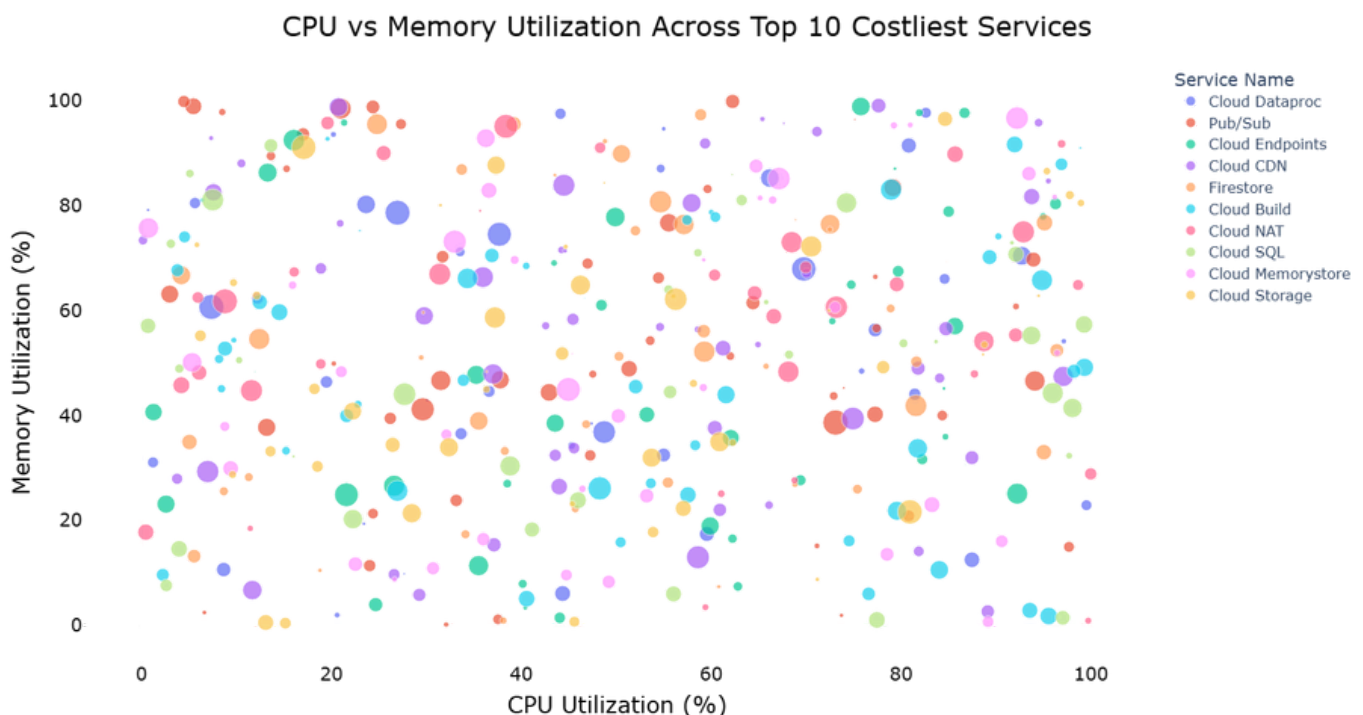
Business Insights

- Regional cost concentration presents an opportunity to evaluate workload distribution strategies; shifting non-latency-sensitive workloads to lower-cost regions could optimize overall cloud expenditure.
- High-spend services across regions should be prioritized for cost optimization initiatives such as rightsizing, committed use discounts, or architectural redesign to reduce recurring cost pressure.

High Cost-Driving Services Exhibit Significant CPU-Memory Imbalance, Indicating Resource Inefficiency Across Workloads

\$18.7M

Total revenue over selected period



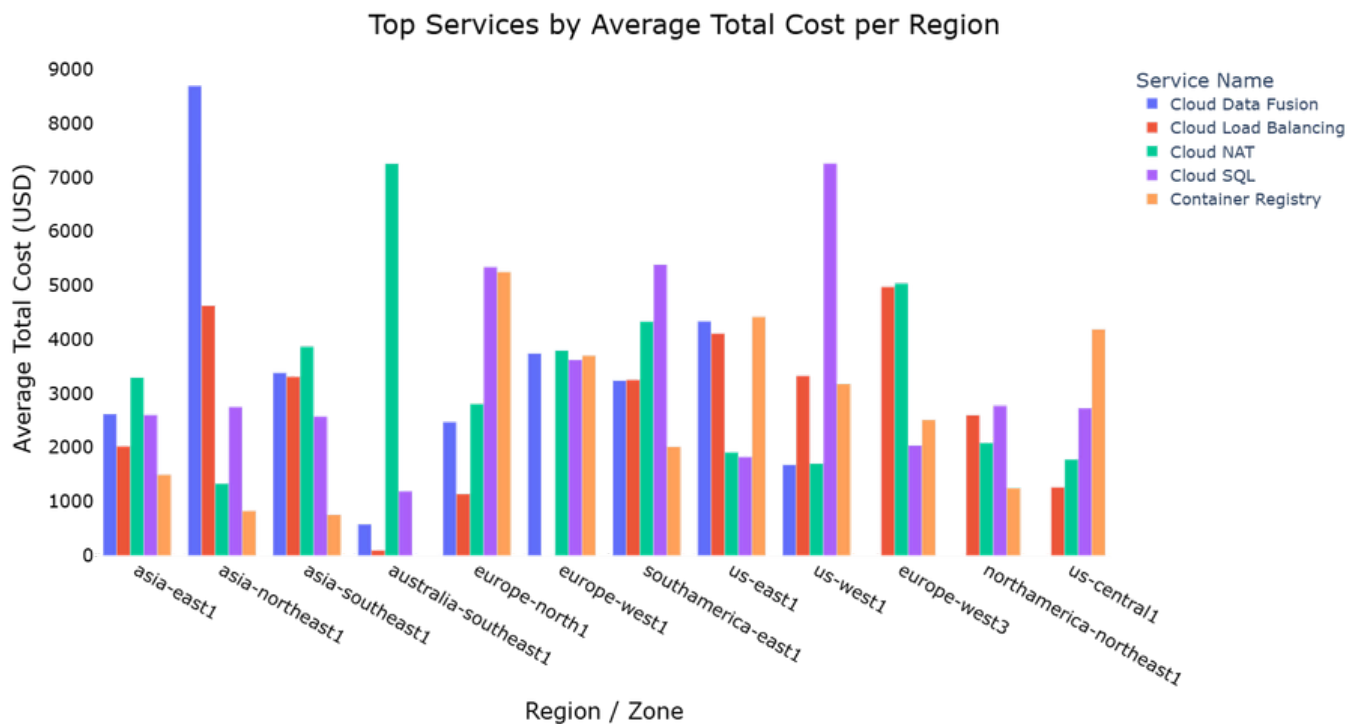
Key observations

- Several high-cost services operate with high CPU but low memory utilization (and vice versa), showing clear resource imbalance across workloads.
- Large cost bubbles are distributed across mid-to-high utilization ranges rather than being concentrated only at peak efficiency levels.

Business Insights

- Imbalanced CPU-memory utilization suggests over-provisioning; rightsizing instances could significantly reduce infrastructure costs without impacting performance.
- High-cost services running below 50% utilization represent immediate optimization opportunities through workload reallocation, autoscaling, or instance resizing.

Cloud SQL and Cloud NAT Dominate Regional Spending, with Significant Cost Concentration in Select Zones

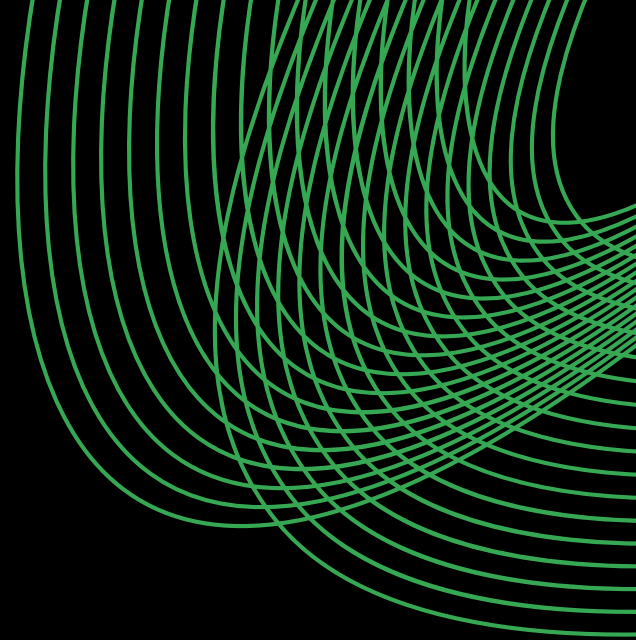


Key observations

- Cloud SQL records the highest average total cost in multiple regions (notably us-west1 and southamerica-east1), indicating strong database-driven spending concentration.
- Certain regions such as asia-northeast1 and australia-southeast1 show sharp cost spikes driven primarily by a single service (Cloud Data Fusion and Cloud NAT respectively).

Business Insights

- Database and network-heavy architectures are major cost drivers; optimizing Cloud SQL configurations and NAT traffic routing could yield substantial savings.
- Regions with single-service cost dominance present targeted optimization opportunities, including workload redistribution or regional cost benchmarking strategies.

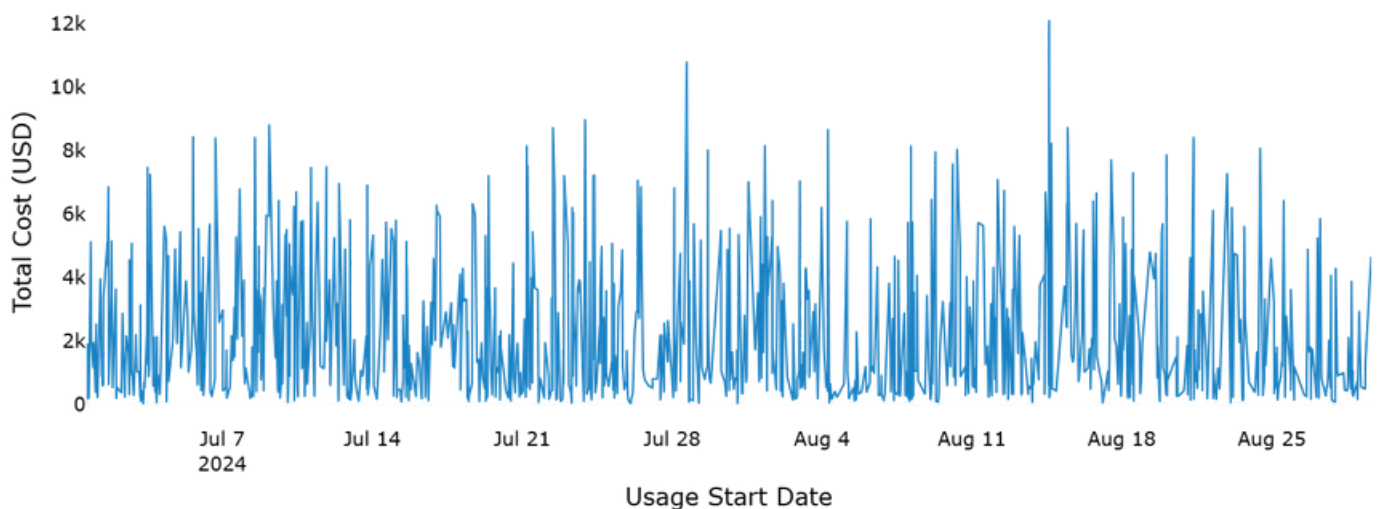


Time Trend Insight



Highly Volatile Daily GCP Spending Indicates Unstable Workload Patterns and Cost Spikes

Cloud Usage Shows Periodic Spikes and Demand Cycles



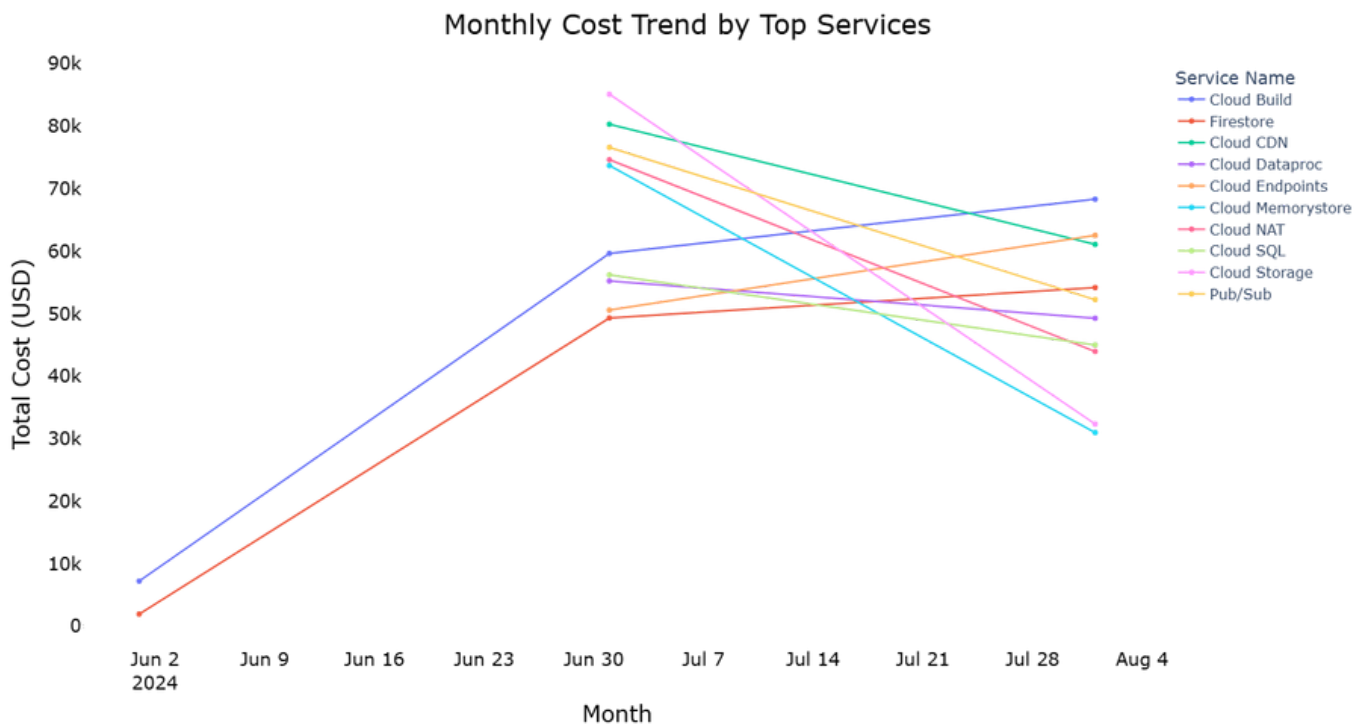
Key observations

- Total daily costs fluctuate sharply throughout July and August, with multiple spikes crossing \$8K–\$12K, indicating irregular usage patterns.
- There is no consistent upward or downward trend, suggesting episodic high-consumption events rather than steady growth.

Business Insights

- Cost volatility signals potential non-optimized scaling, batch processing spikes, or uncontrolled resource provisioning that requires monitoring controls.
- Implementing budget alerts, autoscaling policies, and workload scheduling optimization could reduce unpredictable cost surges and improve financial planning accuracy.

Cloud Build and Cloud Endpoints Show Upward Cost Momentum, While Storage & Dataproc Experience Sharp Cost Decline



Key observations

- Cloud Build and Cloud Endpoints display a consistent upward trend from June to August, indicating growing usage and increasing cost contribution.
- Cloud Storage, Cloud Dataproc, and Cloud Memorystore show a significant cost drop from July to August, suggesting reduced consumption or workload optimization.

Business Insights

- Rising costs in Cloud Build and Cloud Endpoints indicate growing infrastructure dependency, so implementing budget alerts, usage caps, and workload efficiency controls will help prevent uncontrolled cost escalation.
- The sharp decline in Storage and Dataproc costs presents an opportunity to validate whether optimization efforts were successful and replicate similar strategies across other high-cost services.

- **Network Architecture Is a Primary Cost Lever:** Cloud spend is driven more by traffic routing and data transfer patterns than raw compute usage, making network design and egress optimization the highest-impact cost reduction opportunity.
- **Cost Is Highly Concentrated in a Small Resource Subset:** A limited number of high-cost Resource IDs disproportionately influence total expenditure, making targeted deep-dive optimization far more effective than broad cost-cutting efforts.
- **Overprovisioning Is Inflating Infrastructure Spend:** Costs are driven more by allocated instance size and configuration than actual CPU or memory utilization, indicating significant rightsizing and autoscaling opportunities.
- **Resource Configuration Drives More Cost Than Runtime Duration:** High-spec instance types and premium configurations materially increase spend even for shorter workloads, making configuration review more impactful than simply reducing runtime.
- **Utilization Imbalances Signal Immediate Efficiency Gains:** Imbalanced CPU-memory usage and sub-50% utilization across high-cost services highlight clear opportunities for workload tuning, resizing, and instance optimization.



BUSINESS / DEVELOPER TAKEAWAYS

- **Regional Distribution Supports Resilience, Not Cost Efficiency:** The diversified regional footprint likely supports availability and compliance needs, meaning optimization efforts should prioritize service efficiency rather than regional consolidation.
- **Database and Network-Heavy Architectures Require Strategic Review:** Cloud SQL, NAT traffic, and data-heavy services represent structural cost drivers, making architectural redesign and traffic optimization key long-term savings strategies.
- **Cost Volatility Reflects Scaling and Governance Gaps:** Spikes in spend suggest non-optimized scaling or uncontrolled provisioning, reinforcing the need for budget alerts, autoscaling controls, and workload scheduling discipline.
- **Service-Level Trends Reveal Optimization Success Patterns:** The decline in Storage and Dataproc costs demonstrates that targeted optimization works, while rising Cloud Build and Cloud Endpoints costs highlight areas requiring governance and efficiency controls.



CHALLENGES & OPPORTUNITIES

Limitations

- The dataset captures only tracked resource usage and costs within Google Cloud Platform, and may not reflect services or workloads outside the scope of monitoring or unmanaged accounts.
- Observed utilization and spending patterns are limited to the available billing period, potentially underrepresenting seasonal spikes, batch workloads, or long-term trends.
- Timestamped usage and cost events provide a snapshot of activity, which limits the ability to infer long-term architectural efficiency or future demand shifts beyond the recorded timeframe.

Future Work / Opportunities:

- Extend observation windows to analyze longer-term trends in resource usage, cost growth, and service efficiency.
- Incorporate workload metadata or operational logs (e.g., batch jobs, deployment schedules) to better understand usage patterns and performance drivers.
- Compare high-cost vs low-cost resources over time to identify recurring inefficiencies and opportunities for rightsizing or architectural optimization.
- Integrate pricing, tiering, or commitment data to assess the impact of different pricing models on cost predictability and cloud spend optimization.



CONCLUSION

This analysis of the Google Cloud Platform usage and cost dataset provides insight into resource consumption patterns, cost drivers, and infrastructure efficiency across services, regions, and resource types. The findings demonstrate how instance configurations, network traffic, workload provisioning, and service-specific trends influence overall cloud expenditure, while future work could extend this through deeper cost modeling, rightsizing simulations, and scenario analysis for high-cost resources.

This analysis was conducted using Python, Pandas, Matplotlib, Seaborn and Plotly.

Author:

Tejas Jadhav

GitHub: @tejas-jadhav

LinkedIn: Tejas Jadhav

